

Qatar-4b

R-2453

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_221109 · created 2026-08-23T00:11:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459892.6688 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.081 +/- 0.034
Transit depth (Rp/Rs)^2	0.65 +/- 0.56 [%]
Orbital Inclination (inc)	89.59 +/- 2.16
Airmass coefficient 1 (a1)	0.792 +/- 0.019
Airmass coefficient 2 (a2)	0.219 +/- 0.095
Scatter in the residuals of the lightcurve fit is	2.39 %
Best Comparison Star	#1 - [316, 126]
Optimal Method	PSF photometry
Transit Duration (day)	0.0847 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.081 ± 0.034 vs 0.138 (deviation 1.6σ)
Duration fit vs published	0.0847 d vs 0.0892 d (deviation 0.76σ)
Mid-time O-C	-8.6 min
Depth SNR	2.3
Residual scatter	2.39 %

Light curve

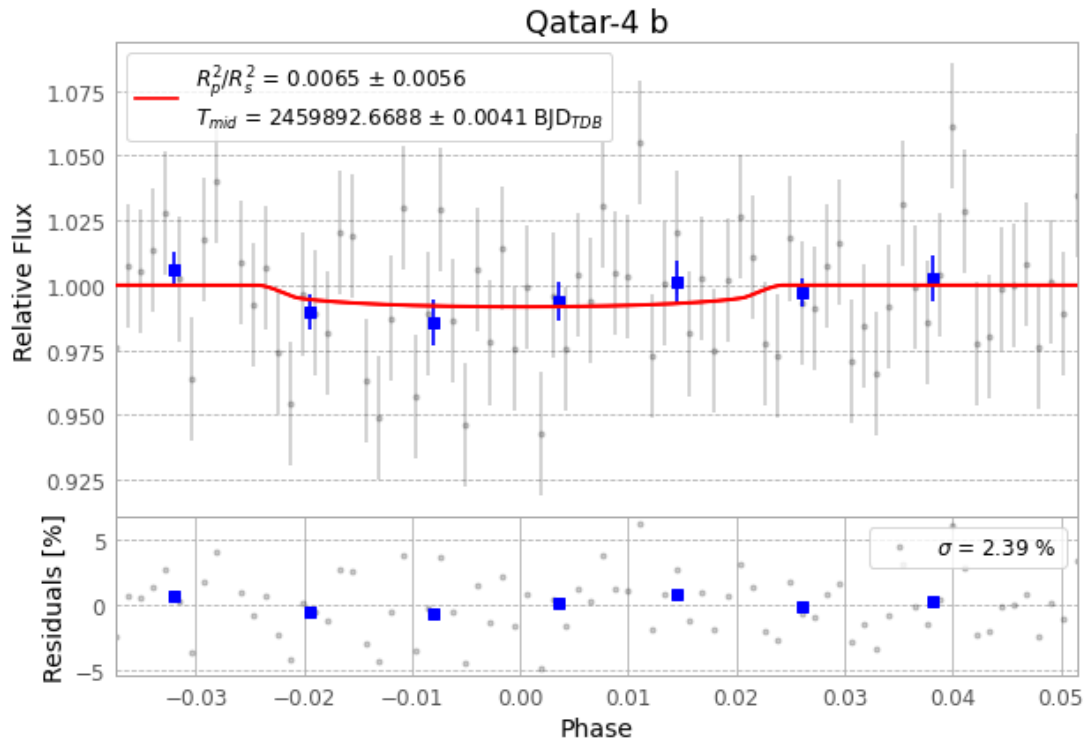


Figure 1 — FinalLightCurve_Qatar-4 b_2022-11-08.png (SHA-256 61be160024b308a5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [459, 135], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 44edfdb99ac92ac...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

78 FITS frames (51 MB), Qatar-4221109021815.FITS → Qatar-4221109060916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-221109.log (53960337a3ed...), FinalLightCurve_Qatar-4 b_2022-11-08.png (61be160024b3...), FinalLightCurve_Qatar-4 b_2022-11-08.csv (e18f3bce6d2a...)

Compute resources

Wall time 132.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 00:11Z.